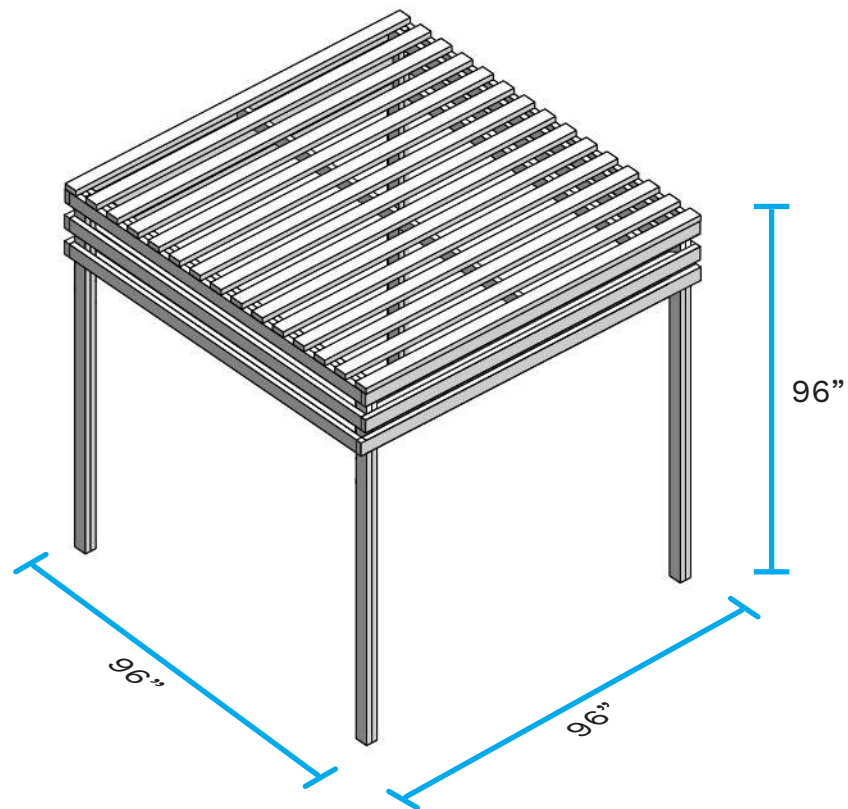




8' SHADE STRUCTURE

ASSEMBLY INSTRUCTIONS

BEFORE YOU BEGIN

1. This document includes recommendations for do-it-yourself (DIY) construction of a [shade structure](#). As a DIY project, the Owner, or the Owner's Contractor, is required to procure, fabricate, assemble, install, and maintain all project materials. These are recommendations only and any material, assembly, and construction must be tailored to specific project requirements.
2. These recommendations are based on common best practices associated with standard DIY home-improvement and/or landscape construction. These best practices may include working knowledge and experience with manual and power tools, hardware, fasteners, lumber, concrete footings, and associated techniques of assembly and construction. Prior working knowledge of these best practices is required so as to avoid harm or personal injury to self or others, or to property.
3. This document includes recommendations for the sequence and assembly of [shade structure](#) components. Locate your [shade structures](#) in your plan you created in the 'Dream' workbook to find the proposed location and dimension of all [shade structures](#). This plan must be tailored to the specific conditions of the project site, including property lines, existing features- such as walls, trees, etc- and any other existing site conditions that may limit or otherwise influence the installation. Use your cost estimation spreadsheet to determine the quantity needed for all your [shade structures](#).
4. Before using any power equipment (such as drills, saws, augers, or other), review all manufacturer safety requirements. While handling or operating any power equipment, follow all manufacturer safety requirements. Do not allow others who have not reviewed and/or follow manufacturer safety requirements to handle or utilize power equipment.
5. The owner is solely responsible for any work required to secure, and comply with, all necessary building and safety codes, permits, and regulatory approvals.
6. These recommendations assume that the project site has been sufficiently prepared for construction. This includes, but is not limited to: identification of utilities that may interfere or otherwise impact installation; the removal of trash, hazards, unwanted structures; or other conditions that may inhibit safe construction.
7. Before construction, review the project site conditions to identify any potential limitations to assembly and installation. This includes determining advance notification by the Owner of utility companies so as to determine the location of all above and below ground utilities.
8. All persons associated with building and construction should sign a volunteer waiver.

LEGAL NOTICE:

The recommendations included herein are based on standard practices of fabrication and construction of garden furnishings. In no event shall Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates be liable for any damages or injury to person or property damages arising out of the use of these recommendations. Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, or its faculty, staff and/or affiliates make no warranties, expressed or implied, and hereby disclaims and negates all other warranties, including without limitation, implied warranties or conditions of merchantability, fitness for a particular purpose, or non-infringement of intellectual property or other violation of rights. Further, Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates do not warrant or make any representations concerning the accuracy, likely results, or reliability of the use of these recommendations. The use of these recommendations constitutes your acknowledgment of the aforementioned.

SHADE STRUCTURE

ASSEMBLY INSTRUCTIONS

COMMON LUMBER SELECTIONS \

WESTERN RED CEDAR:

PRO: NATURALLY WEATHER/ROT RESISTANT LUMBER WITH AESTHETICALLY PLEASING RED TONES AND PLEASANT AROMAS, THAT IS EASY TO WORK WITH. NO PRESERVATION TREATMENTS ARE NECESSARY AS THE SURFACES WILL GRADUALLY DEVELOP A LOVELY GRAY PATINA. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: COST, BE SURE TO SPECIFY 'KNOTTY' (STK) GRADE WHICH IS MUCH MORE AFFORDABLE THAN 'CLEAR' GRADE.

FIR:

PRO: MODERATELY WEATHER/ROT RESISTANT AND STRONG LUMBER. IT IS ALSO EASY TO WORK WITH AT A LOWER COST THAN CEDAR. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: IT WILL NOT STAND THE TEST OF TIME AS WELL AS CEDAR, BUT A GREAT CHOICE ESPECIALLY IF YOU PLAN TO PAINT IT.

PRESSURE TREATED:

PRO: WEATHER/ROT RESISTANT LUMBER AT A LOW COST.

CON: LUMBER IS STILL SUSCEPTIBLE TO CRACKING FROM SUN EXPOSURE, AND OFTEN OF VARIABLE POOR QUALITY AND SATURATED WITH CHEMICAL TREATMENT MAKING IT DIFFICULT TO WORK WITH LEADING TO INCONSISTENT RESULTS. NEEDS 1-4 MONTHS OF AIR DRYING BEFORE PAINTING. NOT EASILY RECYCLABLE AND OFTEN ENDS UP IN LANDFILLS AT THE END OF ITS PRIMARY USE. NOT RECOMMENDED FOR USE IN AREAS WHERE VEGETABLES ARE GROWN.

TIP: ORDER 5-10% MORE LUMBER DEPENDING ON OVERALL QUANTITIES TO ALLOW FOR SOME VARIABLE LUMBER QUALITY AND INCORRECT CUTS.

NOTE: EXTERIOR WOOD TREATMENT CAN PROLONG THE LIFE OF YOUR PARK ELEMENTS. HOWEVER, IT IS NOT NECESSARY TO BRING MANY YEARS OF ENJOYMENT. ANY TREATMENTS MUST BE APPLIED AND MAINTAINED PROPERLY WHICH IS A LABOR INTENSIVE PROCESS. ONE SHOULD DECIDE IF THIS IS PRACTICAL OR NOT FOR THEIR PARK. THE PARK ELEMENTS ARE LOW COST, DURABLE, AND EASY TO BUILD BY DESIGN AND IT MAY BE MORE PRACTICAL TO SIMPLY HAVE THE NEXT GENERATION REBUILD THE ELEMENTS WHEN DEEMED NECESSARY. BELOW ARE RESOURCES FOR CONSIDERATION IF INTERESTED IN EXTERIOR WOOD TREATMENTS.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>

<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>

<https://vermontnaturalcoatings.com/product-category/exterior/>

PARK IN A TRUCK The Landscape Architecture Program + The Lab for Social and Urban Innovation Thomas Jefferson University

WORKING VERSION 1.0

SHADE STRUCTURE \

TOOLS:

10" OR 12" MITER SAW (or appropriate hand saw)

IMPACT DRIVER for self drilling exterior wood screws

DRILL with 1/8" drill bit if needed for pre-drilling holes

5lb HAMMER for J hooks

SANDPAPER 120 GRIT

CARPENTERS SQUARE

SAFETY GOOGLES + GLOVES

6' STEP LADDER

(2) WOOD CLAMPS (OPTIONAL)

MATERIALS:

(268) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS

(4) L BRACKETS

(4) 1/2" x 2" LAG SCREWS

(4) 18"J HOOKS

(4) 18"J HOOKS

CONCRETE DECK BLOCKS (IF NEEDED SEE STEP 7)

LUMBER:

(47) 2" x 4" x 8'-0"

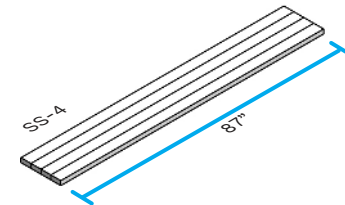
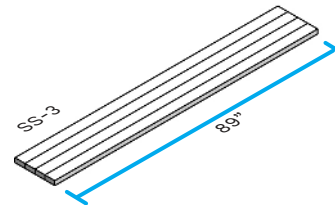
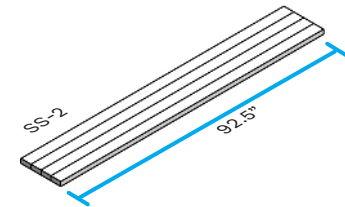
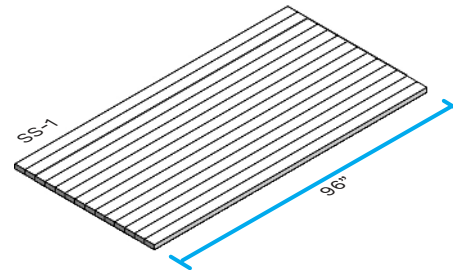
1 STEP

1. MARK, LABEL AND CUT LUMBER IN ORDER OF SS-1 TO SS-4 TO THE FOLLOWING LENGTHS:

NOTE: CHECK EACH PIECE OF LUMBER IMPERFECTIONS SUCH AS CRACKS, WARPING, OR UNEVEN MILLING. DISCARD THESE PORTIONS OF LUMBER.

SHADE STRUCTURE ASSEMBLY

- SS-1: (14)** 2 x 4 x **96"**
LEG LONG, SIDE BRACE,
TOP BRACE, TOP
- SS-2: (4)** 2 x 4 x **92.5"**
LEG SHORT
- SS-3: (4)** 2 x 4 x **89"**
INTERIOR BRACE- SIDE B
(CUT TO SIZE)
- SS-4: (4)*** 2 x 4 x **+/- 87"**
INTERIOR BRACE- SIDE A
(CUT TO SIZE IN STEP 5)



* THESE PIECES ARE A ROUGH ESTIMATE OF SIZE, DO NOT CUT THESE DOWN UNTIL MEASUREMENTS ON SITE HAVE BEEN TAKEN, AS REQUIRED IN STEP 5

TIPS:

- LABEL EACH PIECE OF LUMBER AS YOU ARE MEASURING/CUTTING
- HAVE A PILE OF LUMBER FOR SS-1 TO SS-7
- LIGHTLY SANDING CUT EDGES TO REMOVE SPLINTERS WILL ALSO HELP WITH ASSEMBLY.
- IF BUILDING MULTIPLE STAGES, CREATING A STOP FOR YOUR MITER SAW ENSURES ALL PIECES ARE CUT TO THE SAME LENGTH

2 STEP

NOTE: THE FIRST TWO STEPS CAN BE PERFORMED ON LEVEL GROUND TO ASSIST WITH ASSEMBLY

1. CREATE LEGS

STACK (1) SS-2 TO (1) SS-1 WITH ONE SIDE ALIGNING THE TWO PIECES OF LUMBER, THE OTHER SIDE ALLOWING FOR A 3.5" GAP. THIS WILL FORM THE FIRST LEG OF THE TRELLIS.

2. USE THE PRE-DRILLING GUIDE  AT THE END OF THE PACKET TO PRE-DRILL YOUR SCREW HOLES

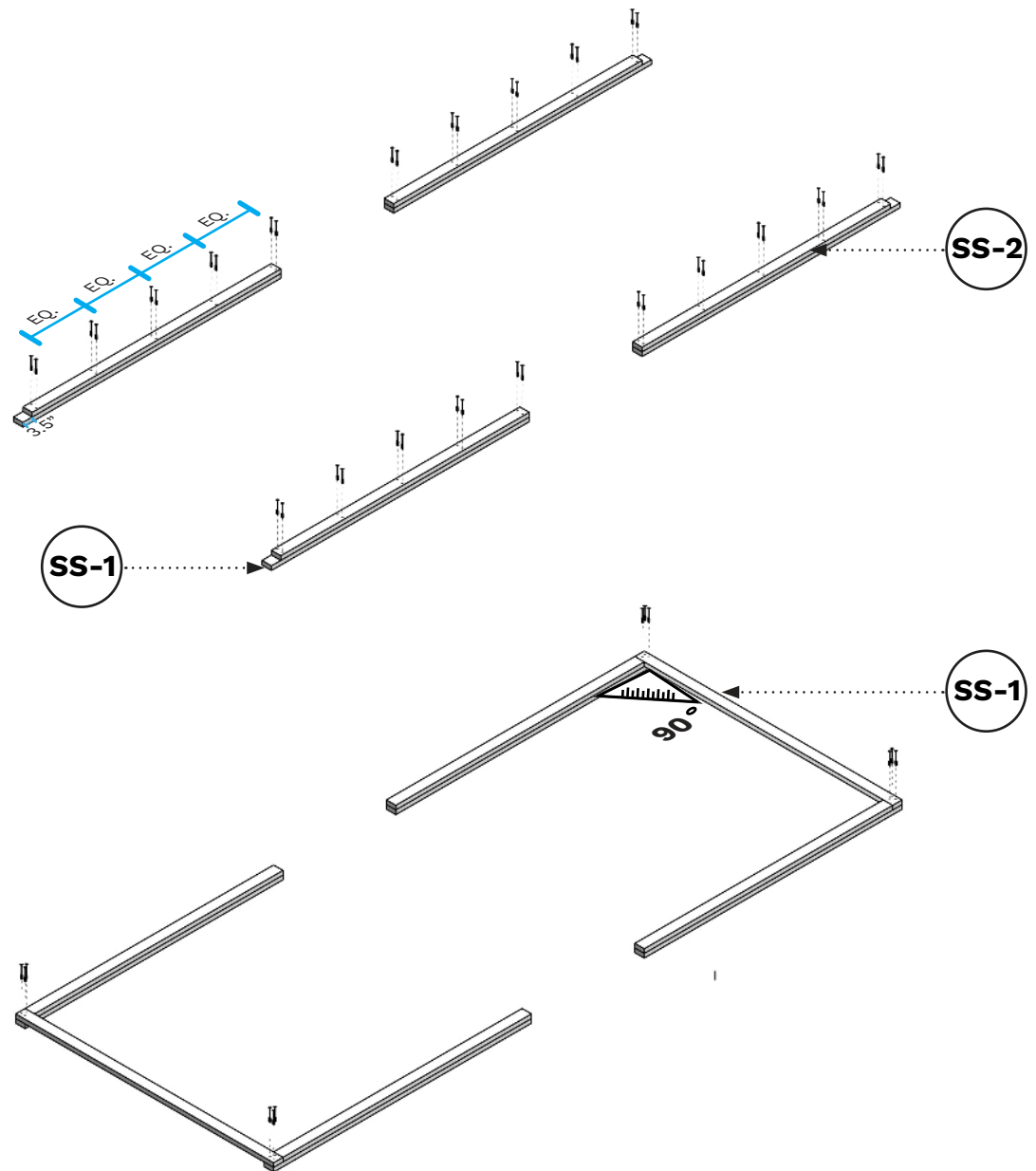
3. SECURE THESE TWO PIECES WITH (2) 2.5" SELF DRIVING EXTERIOR SCREWS AT ONE END OF THE PIECE, AND ANOTHER (2) SCREWS AT THE OTHER END. ADD ANOTHER (2) SCREWS IN THE MIDDLE. THEN ANOTHER SET OF (2) SCREWS EQUAL DISTANT FROM THE END OF THE WOOD AND THE MIDDLE. EACH LEG WILL HAVE A TOTAL OF (10) SELF-DRIVING EXTERIOR SCREWS.

4. REPEAT (3) MORE TIMES WITH ANOTHER (3) SS-2 AND (3) SS-1, AND (30) SELF-DRIVING EXTERIOR SCREWS.

5. CONNECT LEGS

SECURE (1) SS-1 TO THE TOP OF (2) LEGS (SLOTTED INTO THE 3.5" GAP BETWEEN THE SS-2 + SS-1 OF THE LEGS) WITH (8) SELF-DRIVING EXTERIOR SCREWS, (4) PER LEG. REPEAT WITH THE OTHER TWO LEGS.

TIP: ENSURE EACH SIDE STAYS 90 DEGREES BY USING A CARPENTERS SQUARE



3 STEP

NOTE: THE FIRST TWO STEPS CAN BE PERFORMED ON LEVEL GROUND TO ASSIST WITH ASSEMBLY

1. FLIP THE LEGS OVER

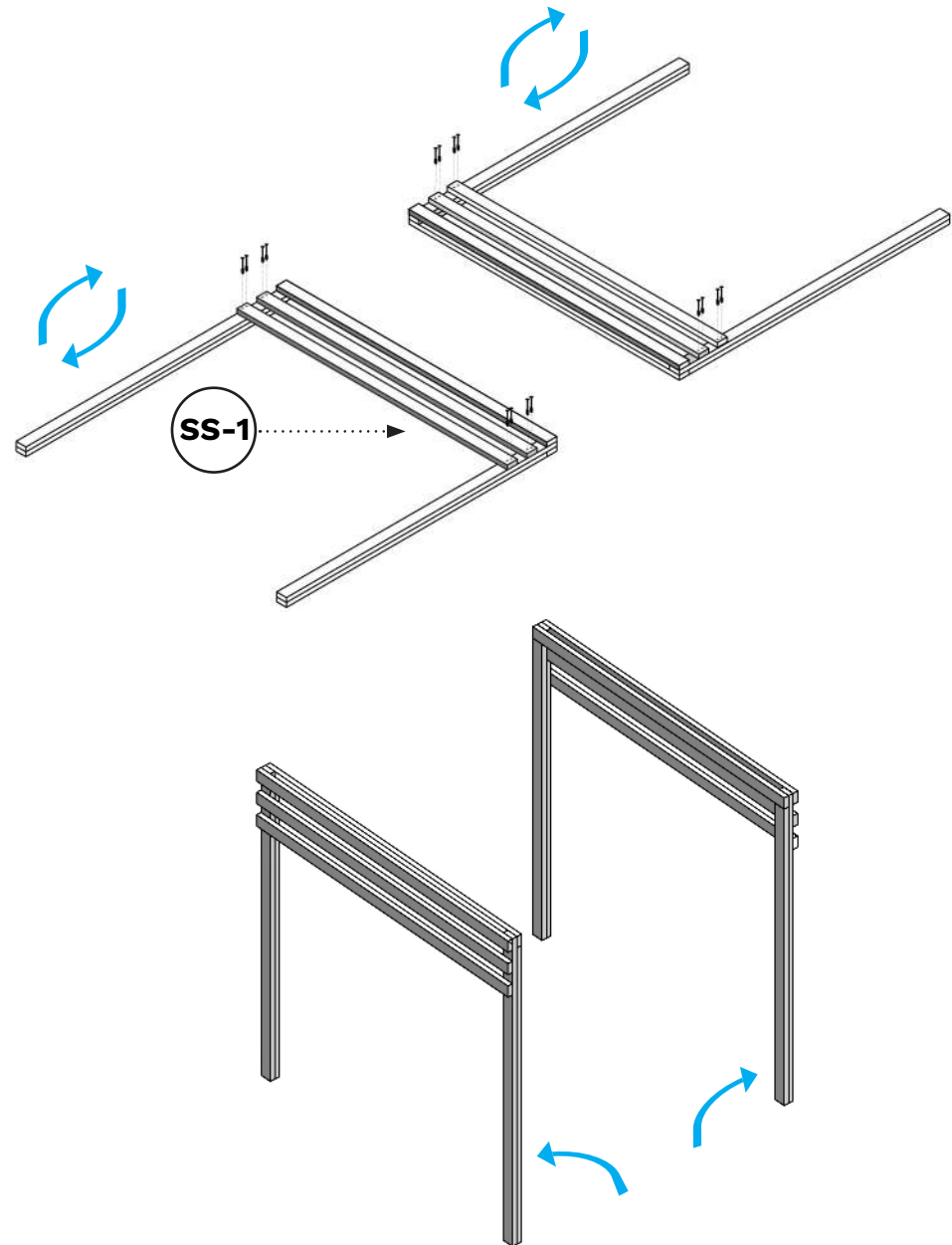
USING (2) SS-1s FOR ONE SET OF LEGS, CREATE BRACES ON ONE SIDE OF YOUR TRELLIS. USE A SPARE 2x4 TO CREATE AN EVEN 3.5" GAP BETWEEN THE TOP LEG BRACE AND THE ADDITIONAL BRACES. EACH BRACE SHOULD USE (4) SELF-DRIVING EXTERIOR SCREWS, (8) TOTAL FOR THIS SIDE.

2. REPEAT ON OTHER LEGS

USING (2) SS-1, SECURE WITH (8) SELF-DRIVING EXTERIOR SCREWS.

3. STAND LEGS UP

TIP: ENSURE EACH SIDE STAYS 90 DEGREES BY USING A CARPENTERS SQUARE



4 STEP

NOTE: THE REMAINING STEPS WILL REQUIRE A LADDER TO ASSIST IN ASSEMBLY

1. ADD TOP CROSS BEAMS

USING (2) SS-1s TO CONNECT EACH OF THE TWO SETS OF LEGS. ENSURE ALL SIDES ARE 90 DEGREES. SECURE EACH CORNER WITH (4) SELF-DRIVING EXTERIOR SCREWS, FOR A TOTAL OF (8) SCREWS FOR THIS STEP.

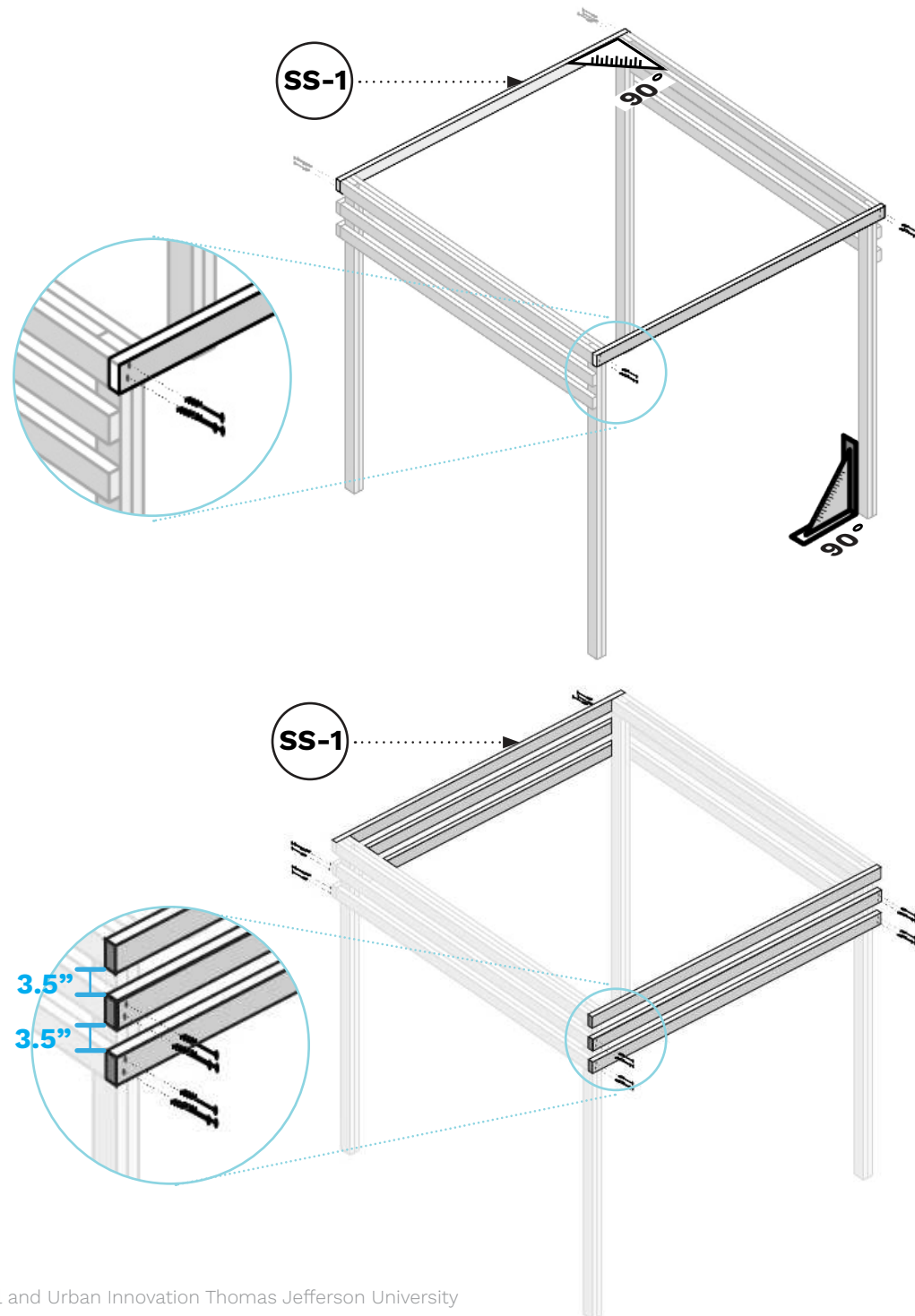
TIP: ENSURE EACH SIDE STAYS 90 DEGREES BY USING A CARPENTERS SQUARE

2. ADD LONG BRACES

USING (2) SS-1s FOR ONE SET OF LEGS, CREATE BRACES ON ONE SIDE OF YOUR TRELLIS. USE A SPARE 2x4 TO CREATE AN EVEN 3.5" GAP BETWEEN THE TOP LEG BRACE AND THE ADDITIONAL BRACES. EACH BRACE SHOULD USE (4) SELF-DRIVING EXTERIOR SCREWS, FOR A TOTAL (16) TOTAL FOR THIS SIDE.

3. REPEAT ON OTHER LEGS

USING (1) SS-1, SECURE WITH (4) SELF-DRIVING EXTERIOR SCREWS.



5 STEP

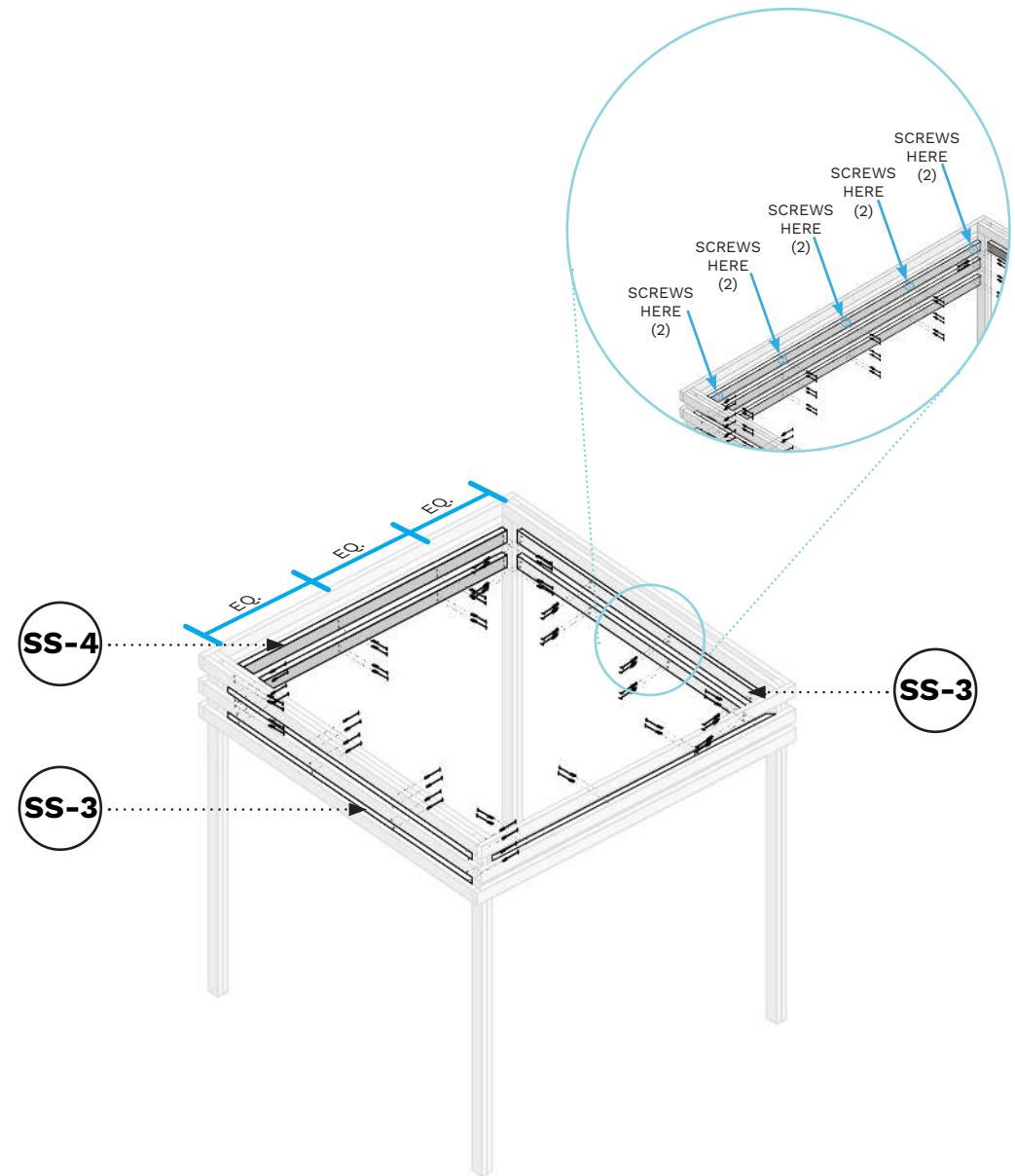
1. MEASURE THE DISTANCE BETWEEN YOUR TWO POSTS WHERE THE SS-4 INTERIOR BRACE SHOULD BE (AS SHOWN). CUT YOUR SS-4 PIECE TO SIZE, ENSURE A SNUG FIT. SECURE TO THE SS-5 PIECE BEHIND IT. REPEAT WITH REMAINING LONG BRACES, FOR A TOTAL OF (4) INTERIOR LONG BRACE PIECES.

2. SECURE IN PLACE WITH (8) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, PLACED IN PAIRS OF (2) AT EQUAL INTERVALS, AS SHOWN. THIS STEP WILL REQUIRE A TOTAL OF (32) SCREWS.

3. MEASURE THE DISTANCE BETWEEN YOUR TWO POSTS WHERE THE SS-3 INTERIOR BRACE SHOULD BE (AS SHOWN) BEFORE SECURING IT IN PLACE. CUT YOUR SS-3 PIECE TO SIZE, ENSURE A SNUG FIT. SECURE TO THE SS-1 PIECE BEHIND IT. REPEAT WITH REMAINING SHORT BRACES FOR A TOTAL OF (4) INTERIOR SHORT BRACE PIECES.

4. SECURE IN PLACE WITH (8) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, PLACED IN PAIRS OF (2) AT EQUAL INTERVALS, AS SHOWN. THIS STEP WILL REQUIRE A TOTAL OF (32) SCREWS.

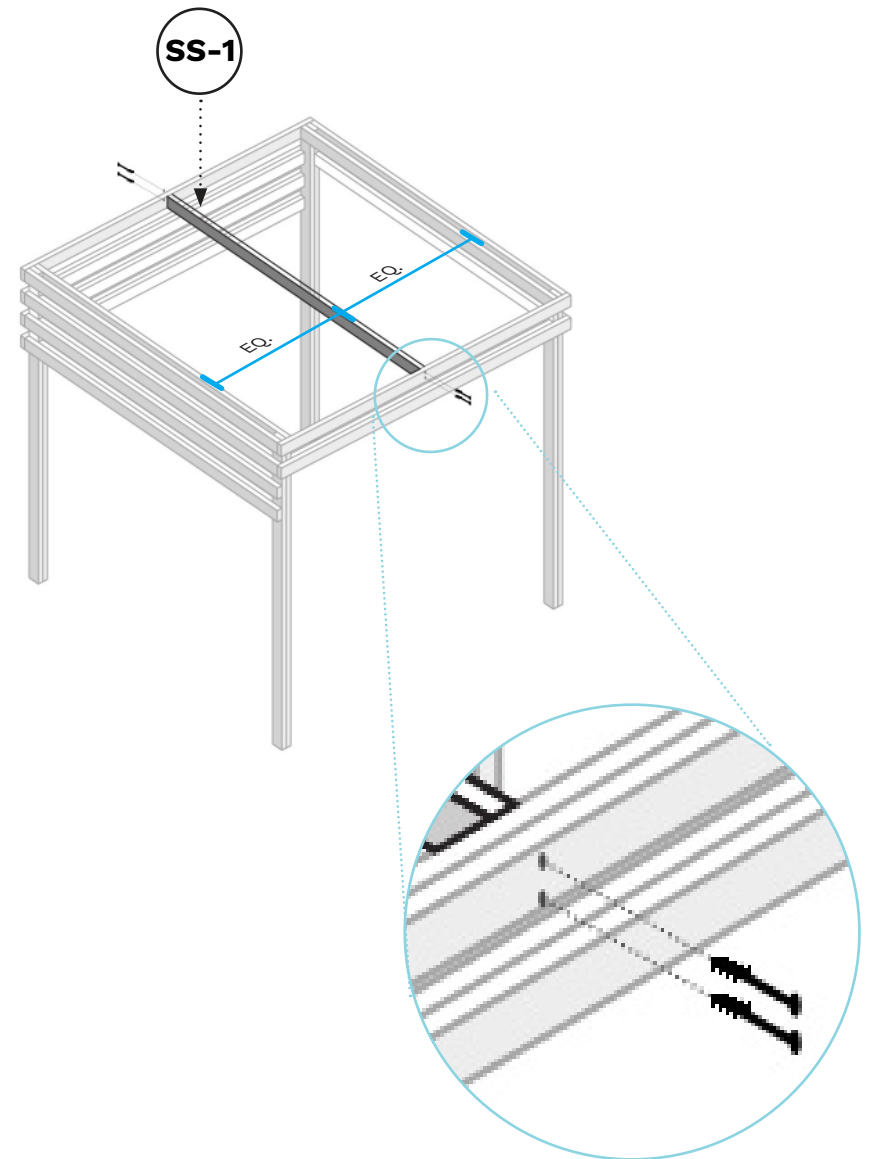
TIP: USE CLAMPS TO HOLD WOOD IN PLACE WHILE SECURING



6 STEP

1. USE (1) SS-1 PIECES TO CREATE THE TOP BRACE SUPPORTS. EQUALLY SPACE AN SS-1 PIECE IN THE CENTER OF THE FRAME, AS SHOWN.
2. SECURE WITH (4) SELF-DRIVING EXTERIOR SCREWS.

TIP: USE CLAMPS TO HOLD WOOD IN PLACE WHILE SECURING



7 STEP

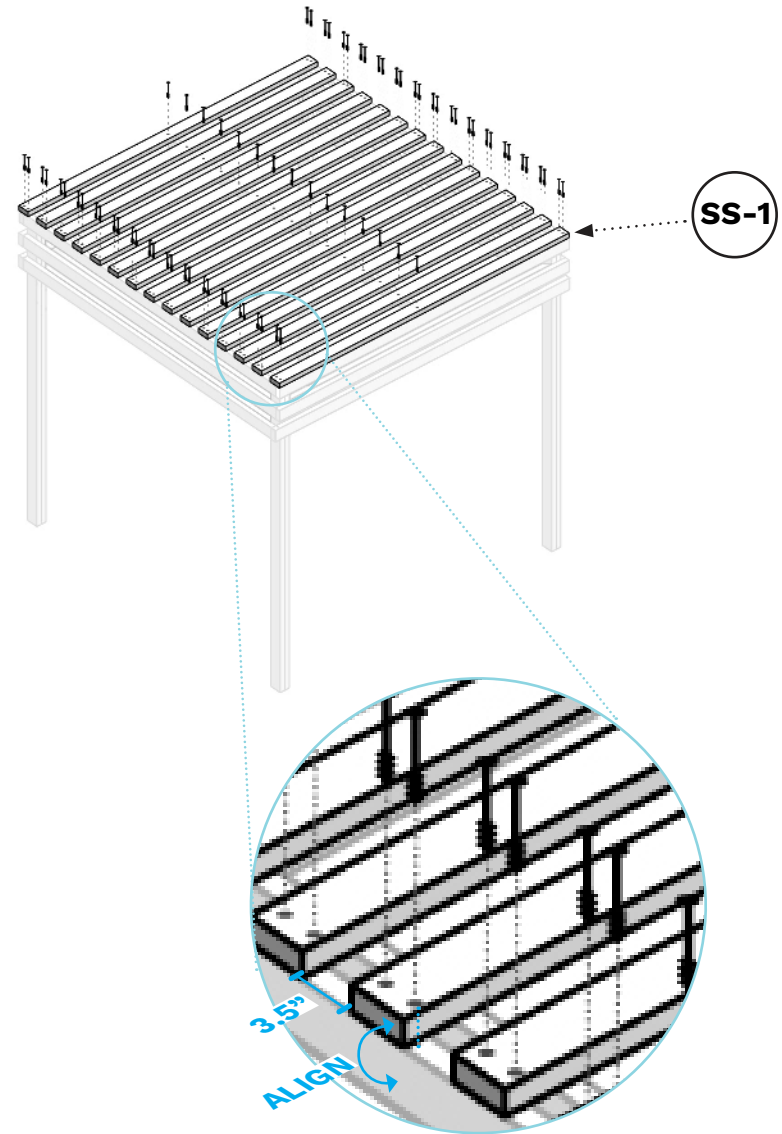
1. USE (15) SS-1 PIECES FOR THE CANOPY. USE A 2x4 AS A SPACER, SECURE EACH SS-5 PIECE FROM ONE SIDE TO THE OTHER (RIGHT TO LEFT) SIDE.

2. SECURE LUMBER BY USING AN IMPACT DRIVER WITH (75) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, AS SHOWN.

3. SECURE REMAINING SS-1 PIECES USING AN IMPACT DRIVER WITH (70) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

4. ALIGN THE LAST PIECE WITH THE FRAME TO ENSURE NO OVERHANG, LETTING THE LAST PIECE BE LESS THAN 3.5" IF NEEDED.

5. LIGHTLY SAND THE ROUGH EDGES OF THE SHADE STRUCTURE.



8 STEP

1. PLACE (4) L BRACKETS ON THE INSIDE OF EACH POST AND SECURE WITH (4) 1/2" x 2" LAG SCREWS.

2. PLACE (4) 18" J HOOKS THROUGH THE BOTTOM HOLE OF EACH L BRACKET AND HAMMER INTO GROUND WITH 5LB HAMMER.



L BRACKETS

NOTE: WHEN SHADE STRUCTURE IS ADJACENT TO GABION BENCH WITH WOOD TOP, IT IS RECCOMENDED TO USE AN L-BRACKETS AND LAG SCREW TO SECURE THE TRELLIS LEGS TO THE GABION BENCH

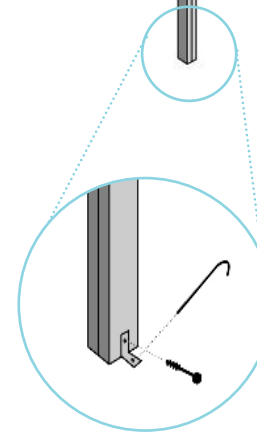
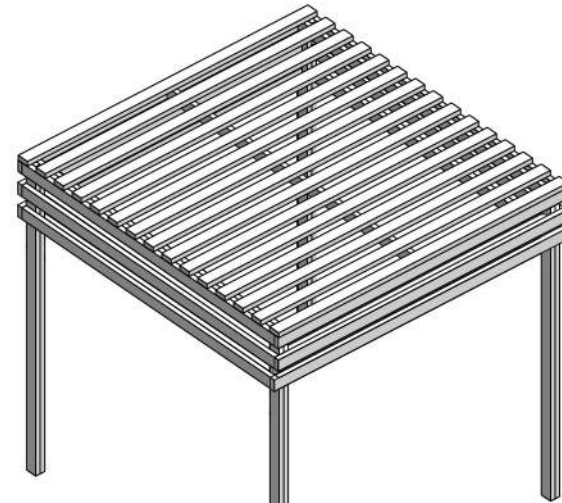


18" J HOOKS



CONCRETE DECK BLOCKS

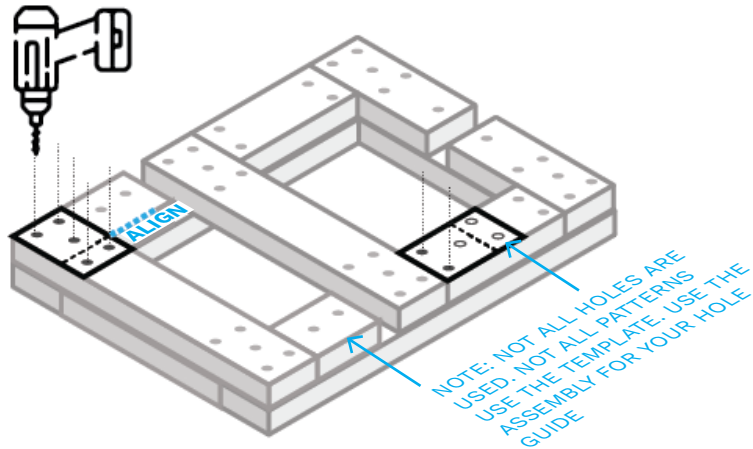
NOTE: WHEN SECURING POST WITHIN PLANTING BEDS, USE A CONCRETE DECK BLOCK TO SET POST ABOVE SOIL AND TO PROVIDE STABLE SUPPORT WHILE MINIMIZING WOOD ROT. BLOCKS WILL HAVE TO BE DUG INTO THE SOIL ACCORDINGLY TO PROVIDE LEVEL SURFACES WITH OTHER POSTS.



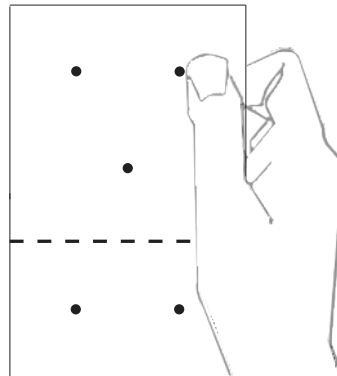


1. CUT OUT PRE-DRILLING HOLE GUIDE ALONG DOTTED LINES

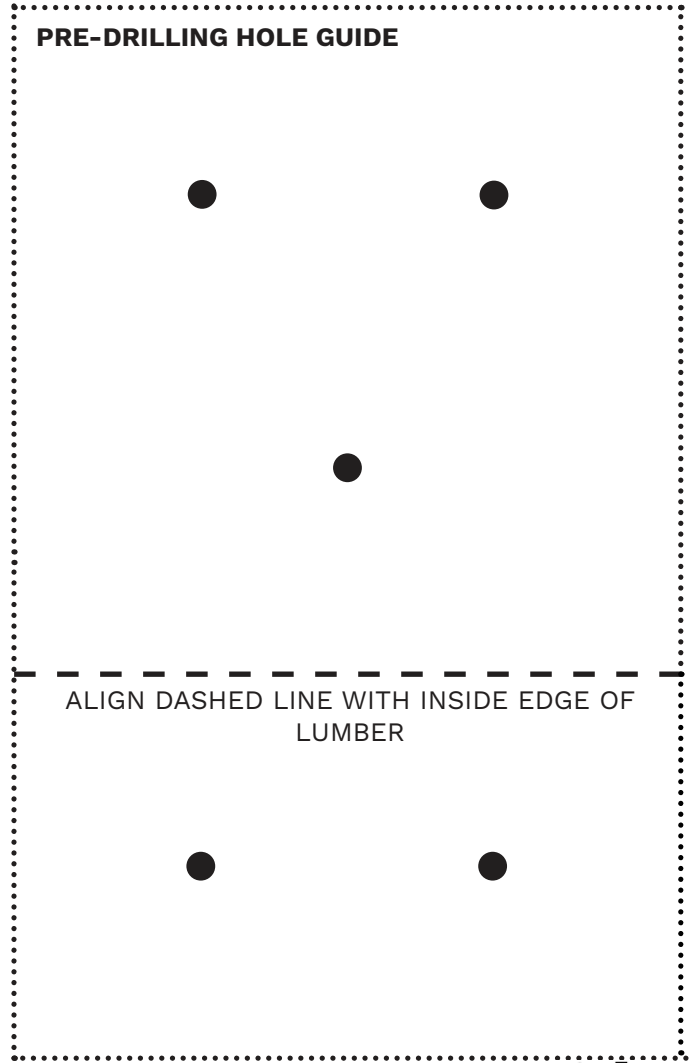
2. WHEN DRILLING, ENSURE THAT YOU PROPERLY **ALIGN THE EDGE OF THE TEMPLATE WITH OUTSIDE EDGE OF THE LUMBER** BEFORE PRE-DRILLING



3. ENSURE ALL HOLES ARE ABOUT A **THUMB IN WIDTH FROM THE EDGE (~1")** EVEN HOLES THAT DON'T USE THIS TEMPLATE



PRE-DRILLING HOLE GUIDE



CUT ALONG THE DOTTED LINES

